

birdclef-2026

Private 9th | Public 2nd solution - Fight against randomness

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Overview

This was my first acoustic competition, and I learned an enormous amount from this community - Mel spectrograms, the STFT, frame- vs clip-level supervision, mostly by reading your notebooks and discussions. Huge thanks to the organizers, the Cornell Lab, Kaggle, and everyone who shared.

If this competition was about one thing for me, it was **fighting randomness**. The metric is noisy - local validation swings $\sim \pm 0.02$ between seeds, the private LB $\sim \pm 0.005$ — so most "improvements" are part signal, part luck, and the real job was telling them apart: validating where it counted, re-running across seeds, and keeping only what survived. Below are all the key elements that gradually helped me climbed up on the public leaderboard, some turned out to work on private as well, some are more or less neutral.

1. Backbone

I used last year's (2025) **Xeno-Canto pretrained EfficientNetV2-S** backbone — credit to [Vladimir Sydorskyy](#), whose [2025 pretrained backbones](#) I built on. This domain-matched pretrain mattered more than anything architectural.

I also self-pretrained a few alternative backbones (hgnetv2_b4, regnety032, seresnext26t) to diversify the ensemble - none worked. eca_nfnet came closest and I kept it for diversity, but on the private LB it was just as weak.

The diversity I imagined wasn't real. EfficientNet was the king.

I also gave up on mel-setup tweaks (multi-mel, fmin sweeps, finer hop - all useless). The mel front-end is coupled to the pretrain, so I kept a single fixed config: `sr=32000, n_mels=128, n_fft=2048, win=2048, hop=512, fmin=20, fmax=16000`

2. Training

CV strategy

- **4 folds** Split: focal = StratifiedKFold by `primary_label`, labeled soundscapes = GroupKFold by filename (fixed split for reproducibility).
- **Validated on soundscapes only**, average macro-AUC over folds as a reference metric - its LB correlation is not ideal.

Basic training params

- AdamW, lr `5e-4`, weight_decay `1e-4`, batch `64`.
- **flat-cosine schedule**, 30 epochs total = 2 warmup + 20 flat + 8 cosine.
- **EMA** (decay 0.999, tracked from epoch 3, used for both validation and the saved weights). EMA stabilized training a lot - it noticeably smoothed the metric curves.
- **BCE Loss** is applied to the **clip-level** logits and the **frame-level** (frame-max) logits.

- **min_sample = 100** (upsample each rare focal class to 100 segments). It gives a real, consistent boost to public LB score.

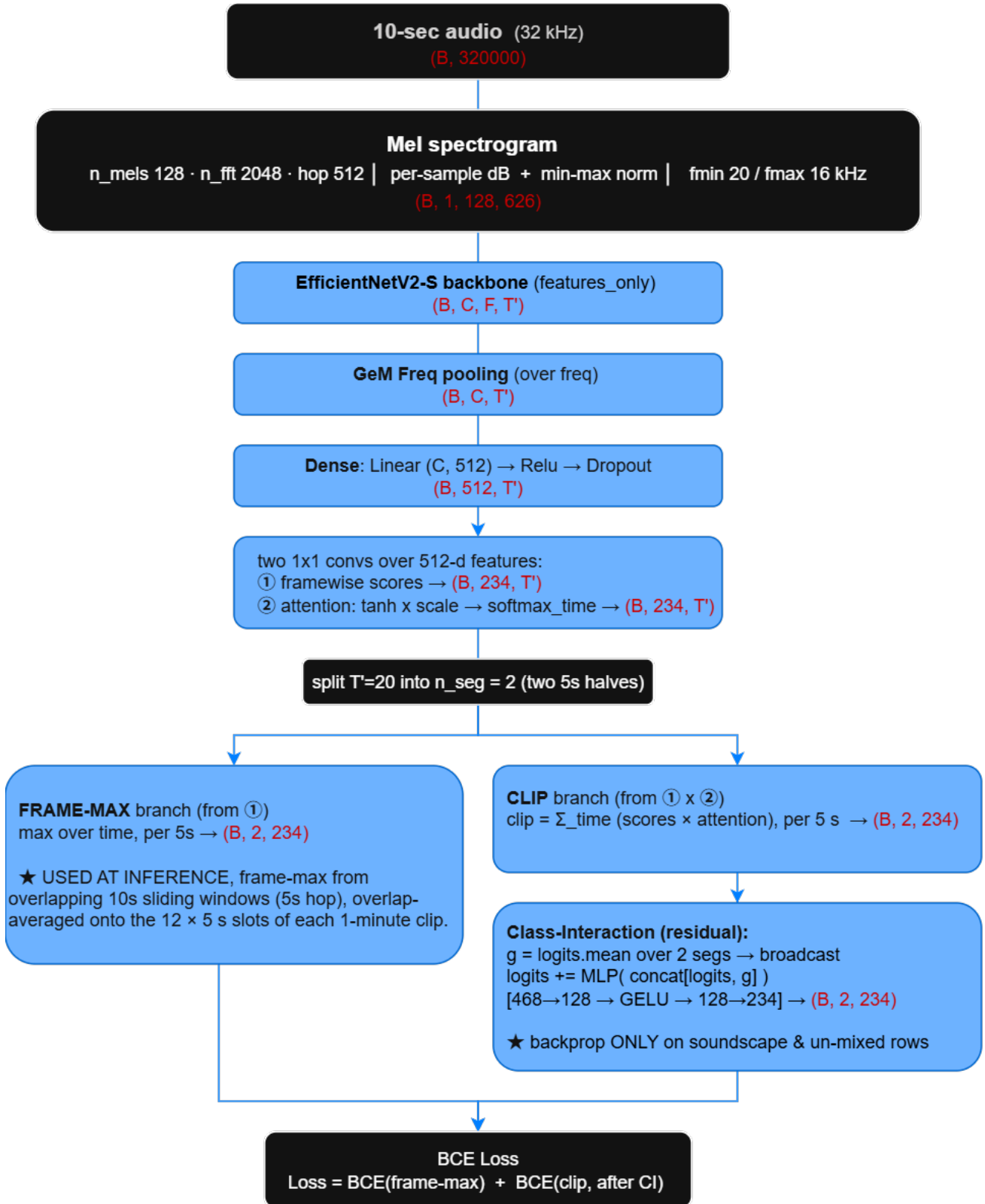
MixUp was the other key knob that bridges the **focal → soundscape domain gap**. Due to the fact that almost all training data is focal (clean, mostly single-species Xeno-Canto clips) and that only a tiny set of soundscapes is available, I mix focal clips of each batch in pair to synthesize soundscape-like, multi-species example. So a model trained mostly on focal transfers to soundscape prediction. Two design choices make this work (MixProb=0.5, MixAlpha=0.4, n_max=2):

- **Mix on the raw waveform, before the mel** (each clip absmax-normalized first).
- **Asymmetric roles for focal vs. the rare soundscape.** Only focal clips get a partner mixed in; the **labeled-soundscape segments are kept clean** (never mixed on the left, `perm[i]=i`), so the scarce real-soundscape supervision stays a pure anchor — but they may serve as partners blended into focal clips. `Beta(0.4)` is U-shaped, so most mixes are one loud + one faint clip, mimicking a soundscape's foreground bird over background calls.

```
lam = Beta(0.4, 0.4).sample((B,))          # U-shaped: usually one loud + one faint
# eligible = focal clips only; partner may be focal OR a (rare) labeled-soundscape segment
perm = partner_index(eligible, exclude=unlabeled_pool) # labeled-SC kept clean on the left
(perm[i]=i)
wav = lam[:,None]*wav + (1-lam)[:,None]*wav[perm]    # raw-waveform mix, before the mel
label = lam[:,None]*label + (1-lam)[:,None]*label[perm] # multi-hot label mix
# only p=0.5 of samples are actually mixed
```

3. Architecture

All my models follow a standard SED stack (EfficientNetV2-S → GeMFreq pooling → attention head → clip + frame logits), at **10s windows, 2 sub-segments**. On top of that, I added a **class-interaction module**: a small residual class×class refinement (Linear→GELU→Linear, zero-init) on the clip-wise logits.



The **class-interaction** module is designed to learn only the real Pantanal species co-occurrence (who actually shows up together in this wetland). It only gets gradients from **un-mixed labeled soundscape samples**. Focal data and mixup'd samples skip it. It starts with zero-init and gradually picks up that clean prior.

Crucially, **class-interaction and the 10s/2 window mostly pay off together**. On the public LB, class-interaction at 5s/1 (exp_B) stays flat at **0.939**, but combined with 10s/2 (exp_D) they jump to **0.952**.

```
# the CI MLP now takes 2*n_cls inputs: [own-segment logits, global mean]
self.ci = nn.Sequential(nn.Linear(2*n_cls, 128), nn.GELU(), nn.Linear(128, n_cls))

def apply_ci_pool(logits, mask):
    # logits: (B, n_seg, n_cls); mask =
    is_soundscape & is_unmixed
    if mask.any():
        # ONLY these rows get refined (and only they
        backprop into ci)
        g = logits.mean(dim=1, keepdim=True).expand(-1, n_seg, -1) # broadcast to each seg
        logits = logits.clone()
        logits[mask] = logits[mask] + self.ci(torch.cat([logits[mask], g[mask]], dim=-1)) # each
        seg sees [self, global]
    return logits
```

I also tried longer durations (15s, 20s), but 10s was the sweet spot — going longer gave nothing.

4. External data

I pulled extra samples for the 234 target species from public sources (Xeno-Canto / iNat), capped at **200 per species**, added them to training (validation untouched), and retrained the same pipeline. This further pushed the public LB up to **0.958**, all without any pseudo-label distillation.

5. Post-processing at inference

I only used two post-processing methods:

- **Taxon-aware smoothing:** texture taxa (Insecta/Amphibia, continuous calls) get mean-smoothed; event taxa (Aves/Mammalia/Reptilia, transient calls) get local-max + a small adaptive shift, so neighbors are cleaned up **without flattening peak confidence**:

```
out[:, texture] = (1-a_tex)*p[:, texture] + a_tex*neighbor_mean(p[:, texture]) # a_tex ~ 0.35
out[:, event] = (1-a_evt)*p[:, event] + a_evt*neighbor_localmax(p[:, event]) # a_evt ~ 0.15
```

- **File-post** (credit: [BirdCLEF 2025 2nd place](#)): each segment's per-species prob is reinforced by that species' top prob across the whole file, boosting species confidently present somewhere in the 1-minute clip:

```
probs = probs.reshape(n_files, 12, n_cls) # 12 five-second slots per 1-min file
file_top = probs.max(axis=1, keepdims=True) # per-file, per-species top prob
probs = probs * file_top # reinforce species confident somewhere in the file
```

A hard-learned warning: a **site/hour prior blend** helped early (when LB was below ~0.95) but started hurting later. It was quietly **contaminating my experiments** (every comparison ran on top of a post-processing step that had turned net-negative). I only caught it very late, then removed it completely ever since.

★ Incremental ablation

The table below summarizes the key experiments — an incremental ablation over the components from sections 1–4, all sharing the same post-processing from section 5. Because the local validation set is small and the metric is noisy, I ran each config at several random seeds and report the one with the best public LB.

exp	dur/n_seg	recipe	public LB	private LB
exp_A	5s/1	baseline	0.939	0.944
exp_B	5s/1	baseline + class-interaction	0.939	0.947
exp_C	10s/2	baseline + 10s	0.946	0.948
exp_D	10s/2	baseline + 10s + class-interaction	0.952	0.950
exp_E	10s/2	baseline + 10s + class-interaction + external data (cap 200)	0.958	0.955

6. Noisy Student

The next step is to exploit unlabeled soundscape, starting by using the best raw models trained in previous steps. Each round of Noisy-Student pseudo-labeling went like this: take the current best ensemble, extract soft labels (sharpened with a **power of 1.3–1.4** to suppress noise), feed them into the next round as MixUp targets - training - then submit, keep the best, and regenerate. After ~2 rounds, a single (multi-fold) model reached **public LB of ~0.96**.

It reliably added ~0.01 on the public LB. On private LB the effect is a bit unclear: pseudo-labels mostly held private **stable around 0.95** across rounds (from my best raw model, exp_D at 0.950, up to 0.951–0.953), slipping only in the already-strong external-data round. Given how noisy this private metric is (the seed sweep above swings $\sim \pm 0.005$), holding steady at ~0.95 reads as a mild **positive**, not pure illusion.

noisy student iteration from exp_D:

exp	public LB	private LB	note
exp_D	0.952	0.950	raw
exp_D+	0.962	0.953	trained with pseudo labels (round 3)
exp_D++	0.963	0.951	trained with pseudo labels (round 5)

noisy student iteration from exp_E (trained with more external data) :

exp	public LB	private LB	note
exp_E	0.958	0.955	raw
exp_E+	0.962	0.954	trained with pseudo labels (round 1)
exp_E++	0.962	0.950	trained with pseudo labels (round 2)

7. Multi-model / multi-backbone re-weighting

My two actual submissions are elaborate pseudo-label ensembles of 4~6 group of models of different backbones (effnet & ecanet). It did help to boost the private LB, but not by much:

sub	public LB	private LB
v1	0.967	0.956
v2	0.966	0.956

Side story: After the competition, I tried a **dead-simple ensemble of just the two raw EfficientNet models (exp_C + exp_D)** - no alternative backbones, no pseudo-labels - and it scored **higher on private than either of my real submissions**:

ensemble	public LB	private LB
exp_D + exp_E (raw)	0.962	0.958

Two raw models beat every complex ensemble I built - including both submissions. Maybe that's just the real world - a pile of clever operations you're proud of, beaten by a few simple decisions.

Closing

Fighting against randomness, the surest way to tilt the odds is also the least elegant: **brute force** - more experiments, more seeds, more comparisons. I missed some good ideas this time (Perch distillation especially — it turned out strong on private), but no regrets: I gave it everything, and it was a blast.

Training & inference code: <https://github.com/yannan90/birdclef-2026-private-9th-public-2nd-solution>

Supplemental Q&A

Antoine Masq · 2026-06-07T07:48:25.410000

First of all congratulations for your solo gold! I really like the idea of the class-interaction module, have you done any ablation experiments on it? I'd be curious if this could be extended to learn the priors for the site, time and dates

Yannan Chen · 2026-06-08T03:26:41.703000

Hi @antoineymasq,

Sorry for the late reply. The CI module is indeed a interesting component I found, and I did run ablations for it. However, some of the results were affected by the improper prior (Site x Hour) post-processing issue mentioned in my write-up, so I'm currently rerunning them.

I'm also testing multiple seeds since there seems to be some randomness. So far the results look good, but a few runs are still pending. I'll update the write-up, together with a fix to the CI module definition that works together with the 10s duration setting, once everything is complete.

Will keep you posted.

Kurise · 2026-06-08T09:52:42.463000

congratulations for your gold medal! May I ask how much improvement you achieved by using last year's backbone?

Yannan Chen · 2026-06-08T13:37:32.927000

Thanks for the question. the pretrained backbone was very helpful, I would say at least +0.01 gain for public LB. at the earlier stages, I was struggling around 0.92 if training from scratch, comparing to 0.93~0.94 after using it.

Yannan Chen · 2026-06-09T02:43:41.777000

Hi @antoinemasq, I updated my notes with more experiments results. Unfortunately, the metrics are extremely unstable & noisy across different seeds. I guess this is mostly due to the small size of the local labeled soundscape, which couldn't provide enough signals for both validation and CI module training. I couldn't be confident to tell you the CI module work for sure. What I can tell you is that in my experiment, 10s duration works better than 5s. And that all my best models are trained with the CI module.

Regarding your idea to learn the priors of site, time, and dates, I'm very hesitate. Mostly because the local labeled soundscapes are so few and cannot form enough true knowledge to learn.

chenwenqiang_001 · 2026-06-13T02:11:10.797000

congratulations, I almost completely According to your description, I have almost fully reproduced your pipeline, but the public leaderboard score is very low. The mel-spectrogram parameters are: sr=32000, n_mels=128, n_fft=2048, win=2048, hop=512, fmin=20, fmax=16000, and the mel-spectrogram size is (128, 313). For the 4-fold split: the focal set uses StratifiedKFold based on primary_label, and the labeled soundscapes set uses GroupKFold based on filename. Then, for each fold, the focal and labeled soundscapes data are combined. I only use fold2 for training, upsample classes with fewer than 100 samples to 100, and follow your training strategy exactly. Finally, with the single fold (fold2) I only achieve 0.87 on the public leaderboard. If it's convenient for you, could you help me check where the mistake might be? Thank you.

Yannan Chen · 2026-06-13T19:11:46.820000

Hey, I never submit single fold at a time. I always submit ensemble of 4 folds together. Even with that, the LB score is still very unstable. I think noisiness is the nature of this competition. Also make sure you use the pretrained weights I mentioned.

chenwenqiang_001 · 2026-06-15T02:13:58.877000

Thank you for your reply. If it is convenient for you, could you answer a few questions? I completely use the weights you mentioned. The pipeline I reproduced is "exp_A 5s/1 baseline 0.939 0.944", In this pipeline whether you add extra datas that you mentioned above. whether you preprocess the focal samples, some of them are corrupted .maybe that be why I didn't reach your LB score?"

Yannan Chen · 2026-06-15T03:54:45.720000

Hi, I didn't add extra data. but I did fix the corrupted ones by redownloading them from xeno-canto. But they are only very few. I don't think it would hugely impact the performance. Another detail I could think of is that while you are using the pretrained backbone, please remember to turn on the "use_features_only".